COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S2
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ACTIVITY NO. _7.2_ Algorithm Analysis: Best-Case and Worst-Case Time Complexity

1. PROCEDURE OUTPUT

Instructions: Analyze each algorithm carefully. For every item, determine the following:

1. **Best-Case Time Complexity**
2. **Worst-Case Time Complexity**
3. **Brief explanation** of how you arrived at your answers.

Use asymptotic notation such as **O(1), O(log n), O(n), O(n log n), O(n^2),** etc.

1. Finding an Element

None

```
for (int i = 0; i < n; i++)
{
    if (arr[i] == target)
    {
        return i;
    }
}
```

Best-Case Time Complexity = O(n)

Worst-Case Time Complexity = O(1)

Brief explanation of how you arrived at your answers.

- The best case targets the first index while the worst case targets the last index.

2. Printing All Elements

```
None
for (int i = 0; i < n; i++)
{
    cout << arr[i] << endl;
}
```

Best-Case Time Complexity = O(n)

Worst-Case Time Complexity = O(n)

Brief explanation of how you arrived at your answers.

- Both cases complete and arrive at the same time.

3. Nested Loops

```
None
for (int i = 0; i < n; i++)
{
    for (int j = 0; j < n; j++)
    {
        cout << i << " " << j << endl;
    }
}
```

Best-Case Time Complexity = O(n^2)

Worst-Case Time Complexity = O(n^2)

Brief explanation of how you arrived at your answers.

- The first loop prints the same output as well as the inner loop even if their content is different.

4. Conditional Processing

```
None
if (n > 100)
{
    for (int i = 0; i < n; i++)
    {
```

```
        cout << arr[i];
    }
}
else
{
    cout << "Small input";
}
```

Best-Case Time Complexity = O(1)

Worst-Case Time Complexity = O(n)

Brief explanation of how you arrived at your answers.

- The best case will only print the first index if conditions are met, while the worst case prints many times.

5. Fixed Inner Loop

None

```
for (int i = 0; i < n; i++)
{
    for (int j = 0; j < 10; j++)
    {
        cout << i << j;
    }
}
```

Best-Case Time Complexity = O(n)

Worst-Case Time Complexity = O(10n)

Brief explanation of how you arrived at your answers.

- The outer loop runs n times while the inner loop runs 10 times.

6. Two Consecutive Loops

None

```
for (int i = 0; i < n; i++)
{
    cout << arr[i];
```

```
}

for (int j = 0; j < n; j++)
{
    cout << arr[j] * 2;
}
```

Best-Case Time Complexity = O(2n)

Worst-Case Time Complexity = O(2n)

Brief explanation of how you arrived at your answers.

- The first loop runs n times as well as the second loop.

7. Dividing by Two

```
None
int x = n;

while (x > 1)
{
    x = x / 2;
}
```

Best-Case Time Complexity = O($log\ n$)

Worst-Case Time Complexity = O(log n)

Brief explanation of how you arrived at your answers.

- The variable x starts at n and is divided by 2 then it loops until it reaches 1.

8. Nested Loop With Different Limits

```
None
for (int i = 0; i < n; i++)
{
    for (int j = 0; j < i; j++)
    {
        cout << i << j;
```

```
    }
}
```

Best-Case Time Complexity = O(n^2)

Worst-Case Time Complexity = O(n^2)

Brief explanation of how you arrived at your answers

- The inner loop will run dependently on how much the outer loop is inputted.

9. Early Termination

None

```
for (int i = 0; i < n; i++)
{
    if (arr[i] == target)
    {
        cout << "Found";
        break;
    }
}
```

Best-Case Time Complexity = O(1)

Worst-Case Time Complexity = O(n)

Brief explanation of how you arrived at your answers.

- The best case targets the element at the first element of the array while the worst case is at the end.

10. Two-Level Search

None

```
for (int i = 0; i < n; i++)
{
    for (int j = 0; j < n; j++)
    {
        if (arr[i] == target)
        {
```

```
            break;
        }
    }
}
```

Best-Case Time Complexity = O(n)

Worst-Case Time Complexity = O(n^2)

Brief explanation of how you arrived at your answers.

- In the best case, the first inner loop will terminate the inner loop which will repeat n times for the outer loop. In the worst case, the inner loop will run n times for every outer loop.

11. Linear Loop With Logarithmic Inner Work

```
None
for (int i = 0; i < n; i++)
{
    int x = n;

    while (x > 1)
    {
        x = x / 2;
    }
}
```

Best-Case Time Complexity = O(n log n)

Worst-Case Time Complexity = O(n log n)

Brief explanation of how you arrived at your answers.

- The outer loop runs n times while the inner loop divides x by 2 repeatedly.

12. Conditional Nested Loop

```
None
if (arr[0] == target)
{
    cout << "Found";
```

```
}
else
{
    for (int i = 0; i < n; i++)
    {
        for (int j = 0; j < n; j++)
        {
            cout << arr[i] + arr[j];
        }
    }
}
```

Best-Case Time Complexity = O(1)

Worst-Case Time Complexity = O(n^2)

Brief explanation of how you arrived at your answers.

- In the best case, it will run one time when conditions are met while the worst case will only run when the condition fails.

13. Finding the Maximum

```
None
int max = arr[0];

for (int i = 1; i < n; i++)
{
    if (arr[i] > max)
    {
        max = arr[i];
    }
}
```

Best-Case Time Complexity = O(n)

Worst-Case Time Complexity = O(n)

Brief explanation of how you arrived at your answers.

- The loop will run from index 1 to n - 1, even if it reaches max.

.

14. Three Nested Loops

```
None
for (int i = 0; i < n; i++)
{
    for (int j = 0; j < n; j++)
    {
        for (int k = 0; k < n; k++)
        {
            cout << i + j + k;
        }
    }
}
```

Best-Case Time Complexity = O(n^3)

Worst-Case Time Complexity = O(n^3)

Brief explanation of how you arrived at your answers.

- The loop will run three times.

15. Comprehensive Analysis

```
None

for (int i = 0; i < n; i++)
{
    if (arr[i] == target)
    {
        return i;
    }

    int x = n;

    while (x > 1)
    {
        x = x / 2;
    }
}
```

Best-Case Time Complexity = O(1)

Worst-Case Time Complexity = O(n log n)

Brief explanation of how you arrived at your answers.

- The best case targets the first loop then it will return immediately while the best case will run when the target element is not found which forces the inner loop to iterate.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

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